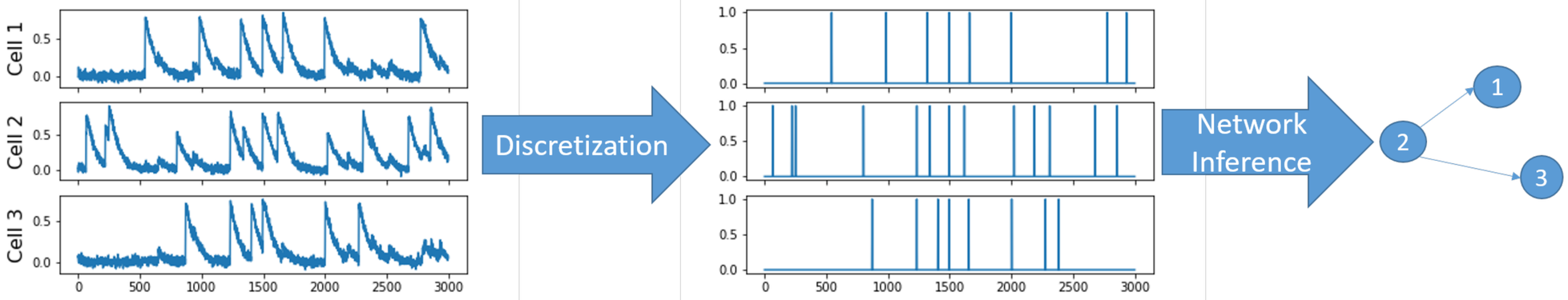
Network Inference from Neural Activation Time Series

A comparative review

George Panagopoulos
University of Houston
https://giorgospanagopoulos.github.io

Introduction

Problem: Infer the underlying connectivity of a neural circuit given the time series of neural activations
Motivation: Understand functional connectivity mechanisms in neural populations and study neural circuits without acute neuroimaging experiments



Unsupervised

Model-free
Two neurons are connected if a similarity metric on their spike trains surpasses a threshold

- Cross Correlation: $\sum_t[(m_t - \overline{m})(n_{t+\tau} - \overline{n})]$.
- Partial Correlation using Σ^{-1} with 80% of variance retained.
- Partial Correlation using Graphical Lasso: $\text{argmin}_{\Theta \geq 0}(tr(\Sigma\Theta) - logdet(\Theta) + \lambda|\Theta|_1)$.

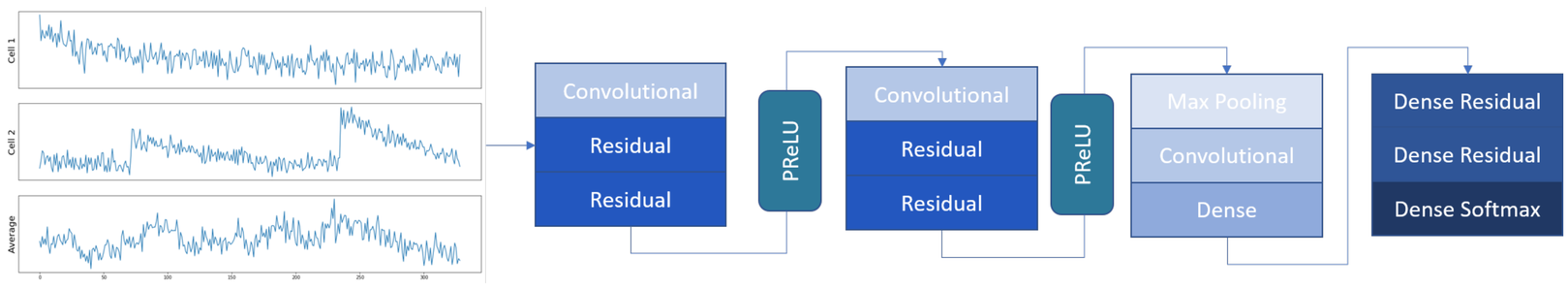
Hawkes Process [2]
Each spike train is modeled as a Poisson process and the whole set is modeled as a Hawkes process

$$p(s_{m,\tau}) = p(s_{m,\tau}|\lambda_m) \times \prod_{t<\tau} \prod_{n \in N} p(s_{t,n}|A_{m,n}W_{m,n}G_{\theta_{m,n}}(t-\tau))$$

Connectivity parameters A & W are estimated using stochastic variational inference

Supervised

Residual Convolutional Neural Network (RCNN)[3]
For each neuron pair, a subset of neural activations is extracted from a random point to serve as input sample to the RCNN. Connected pairs are sampled more to account for class imbalance.



Computing Impulse Responses Using Social Influence Model (CIRUSIM)

1. Count the time spans between a neuron's spike and the immediate spikes of all other neurons
2. Overlook spikes with less than 1 second difference from a spike preceding the current spike
3. For each neuron pair, transform time span series into scores using $e^{1.0/(x)} - 1$ and compute the number of impulses, mean, variance and 95th percentile
4. SVM with RBF kernel and adjusted class weights using the features from step 3

Results

Network Inference Method	AUC %	PRC %	Time (Sec)
Graphical Lasso	83.1	44.2	40
Cross Correlation	77.7	34.9	432
PCA	76.1	30.7	33
Hawkes	72.8	35.5	5588
CIRUSIM	68.1	18.9	3276
RCNN	66	18.4	97[a]

[a]Run in an Amazon g2.2 instance with deep learning Ubuntu AMI

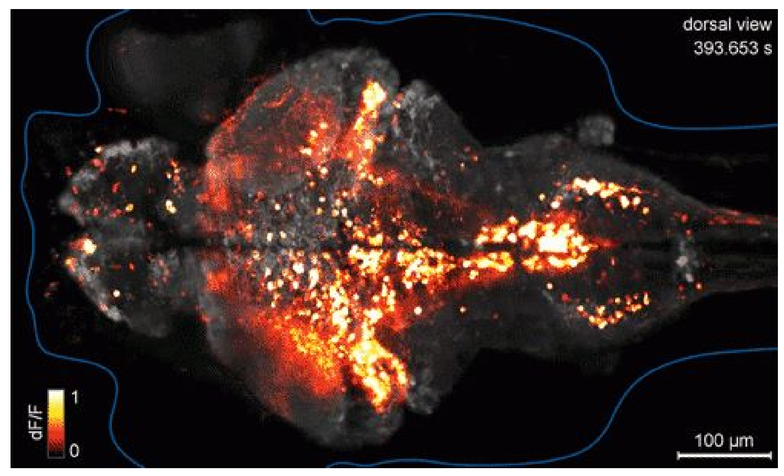
Code

https://github.com/GiorgosPanagopoulos/Network-Inference-From-Neural-Activations

Data

Chalearn Connectomics Challenge [a]

- Simulated fluorescence signals based on larval zebrafish calcium imaging
- 6 networks of 100 neurons
- One hour of recordings in 50Hz
- Ground truth connections



Preprocessing

1. Removal of light scattering
2. Deconvolution using OASIS [1]
3. Discretization with threshold at 0.12

[a]https://www.kaggle.com/c/connectomics

Discussion

- Graphical lasso has the best performance due to partial correlation.
- Supervised methods perform worse due to the leave one network out validation method. CIRUSIM achieves 81% AUC with classic 5-fold cross validation.
- RCNN architecture is built for networks of 1000 neurons

References

[1] Friedrich, Johannes, and Liam Paninski. "Fast active set methods for online spike inference from calcium imaging." *Advances In Neural Information Processing Systems*. 2016.
[2] Linderman, Scott, and Ryan Adams. "Discovering latent network structure in point process data." *International Conference on Machine Learning*. 2014.
[3] Dunn, Timothy, and Peter K. Koo. "Inferring Functional Neural Connectivity With Deep Residual Convolutional Networks." bioRxiv 2017: 141010.